

# Multi-Turn Speech Dialogue

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A curated paper list for Multi-Turn Speech Dialogue.

## Survey

1. "Turn-taking in Conversational Systems and Human-Robot Interaction: A Review". *Gabriel Skantze* Computer Speech & Language 2021. [Paper](#)
2. "A Review of Dialogue Systems: From Trained Monkeys to Stochastic Parrots". *Atharv Singh Patlan et al.* arXiv 2023. [Paper](#)
3. "WavChat: A Survey of Spoken Dialogue Models". *Shengpeng Ji et al.* arXiv 2024. [Paper](#)
4. "From Turn-Taking to Synchronous Dialogue: A Survey of Full-Duplex Spoken Language Models". *Yuxuan Chen and Haoyuan Yu* arXiv 2025. [Paper](#)
5. "A Survey of Recent Advances on Turn-taking Modeling in Spoken Dialogue Systems". *Galo Castillo-López et al.* IWSDS 2025. [Paper](#)
6. "Turn-Taking Modelling in Conversational Systems: A Review of Recent Advances". *Rutherford Agbeshi Patamia et al.* Technologies 2025. [Paper](#)
7. "A Survey of Full-Duplex Spoken Dialogue Systems: Architectural Hierarchy, Interaction Ontology, and Decision State Machine". *Jingyu Lu et al.* arXiv 2026. [Paper](#)

## General Surveys

1. "A Survey of Available Corpora For Building Data-Driven Dialogue Systems: The Journal Version". *Iulian V. Serban et al.* Dialogue & Discourse 2018. [Paper](#)
2. "Neural Approaches to Conversational AI". *Jianfeng Gao et al.* Foundations and Trends in Information Retrieval 2019. [Paper](#)
3. "Recent Neural Methods on Dialogue State Tracking for Task-Oriented Dialogue Systems: A Survey". *Vevake Balaraman et al.* SIGDIAL 2021. [Paper](#)
4. "Who Says What to Whom: A Survey of Multi-Party Conversations". *Jia-Chen Gu et al.* IJCAI 2022. [Paper](#)
5. "Transformers in Speech Processing: A Survey". *Siddique Latif et al.* arXiv 2023. [Paper](#)
6. "A Survey on Recent Advances in LLM-Based Multi-turn Dialogue Systems". *Zihao Yi et al.* arXiv 2024. [Paper](#)
7. "Preference Tuning with Human Feedback on Language, Speech, and Vision Tasks: A Survey". *Genta Indra Winata et al.* arXiv 2024. [Paper](#)
8. "A Survey on Speech Large Language Models". *Jing Peng et al.* arXiv 2024. [Paper](#)
9. "On The Landscape of Spoken Language Models: A Comprehensive Survey". *Siddhant Arora et al.* arXiv 2025. [Paper](#)
10. "Recent Advances in Speech Language Models: A Survey". *Wenqian Cui et al.* ACL 2025. [Paper](#)
11. "Aligning Multimodal LLM with Human Preference: A Survey". *Tao Yu et al.* arXiv 2025. [Paper](#)

## Datasets

1. "Japanese Dialogue Corpus of Information Navigation and Attentive Listening Annotated with Extended ISO-24617-2 Dialogue Act Tags". *Koichiro Yoshino et al.*. LREC 2018. [Paper](#)
2. "Taskmaster-1: Toward a Realistic and Diverse Dialog Dataset". *Bill Byrne et al.*. EMNLP-IJCNLP 2019. [Paper](#)
3. "Alexa in the Wild: Collecting Unconstrained Conversations with a Modern Voice Assistant in a Public Environment". *Benjamin Benk et al.*. LREC 2020. [Paper](#)
4. "HarperValleyBank: A Domain-Specific Spoken Dialog Corpus". *Mike Wu et al.*. arXiv 2020. [Paper](#)
5. "The MSP-Conversation Corpus". *Luz Martinez-Lucas et al.*. Interspeech 2020. [Paper](#)
6. "ASCEND: A Spontaneous Chinese-English Dataset for Code-switching in Multi-turn Conversation". *Holy Lovenia et al.*. arXiv 2021. [Paper](#)
7. "DailyTalk: Spoken Dialogue Dataset for Conversational Text-to-Speech". *Keon Lee et al.* arXiv 2022. [Paper](#)
8. "DOROTHIE: Spoken Dialogue for Handling Unexpected Situations in Interactive Autonomous Driving Agents". *Ziqiao Ma et al.* EMNLP Findings 2022. [Paper](#)
9. "Collection and Analysis of Travel Agency Task Dialogues with Age-Diverse Speakers". *Michimasa Inaba et al.*. LREC 2022. [Paper](#)
10. "Design and Evaluation of the Corpus of Everyday Japanese Conversation". *Hanae Koiso et al.*. LREC 2022. [Paper](#)
11. "SpokenWOZ: A Large-Scale Speech-Text Benchmark for Spoken Task-Oriented Dialogue Agents". *Shuzheng Si et al.* arXiv 2023. [Paper](#)
12. "CALLS: Japanese Empathetic Dialogue Speech Corpus of Complaint Handling and Attentive Listening in Customer Center". *Yuki Saito et al.* Interspeech 2023. [Paper](#)
13. "Advancing Large Language Models to Capture Varied Speaking Styles and Respond Properly in Spoken Conversations (StyleTalk)". *Guan-Ting Lin et al.* ACL 2024. [Paper](#)
14. "Generative Expressive Conversational Speech Synthesis (NCSSD)". *Rui Liu et al.* ACM MM 2024. [Paper](#)
15. "J-CHAT: Japanese Large-scale Spoken Dialogue Corpus for Spoken Dialogue Language Modeling". *Wataru Nakata et al.* arXiv 2024. [Paper](#)
16. "DeepDialogue: A Multi-Turn Emotionally-Rich Spoken Dialogue Dataset". *Alkis Koudounas et al.* arXiv 2025. [Paper](#)
17. "UltraVoice: Scaling Fine-Grained Style-Controlled Speech Conversations for Spoken Dialogue Models". *Wenming Tu et al.* arXiv 2025. [Paper](#)
18. "Toward Conversational Hungarian Speech Recognition: Introducing the BEA-Large and BEA-Dialogue Datasets". *Mate Gedeon et al.* arXiv 2025. [Paper](#)
19. "InteractSpeech: A Speech Dialogue Interaction Corpus for Spoken Dialogue Model". *Yifu Chen et al.* EMNLP Findings 2025. [Paper](#)
20. "Behavior-SD: Behaviorally Aware Spoken Dialogue Generation with Large Language Models". *Sehun Lee et al.* NAACL 2025. [Paper](#)

21. "RealTalk-CN: A Realistic Chinese Speech-Text Dialogue Benchmark With Cross-Modal Interaction Analysis". *Enzhi Wang et al.* arXiv 2025. [Paper](#)
22. "DialogueAgents: A Hybrid Agent-Based Speech Synthesis Framework for Multi-Party Dialogue". *Zheyuan Zhang et al.* arXiv 2025. [Paper](#)
23. "Open-Source Full-Duplex Conversational Datasets for Natural and Interactive Speech Synthesis". *Rui Liu et al.* arXiv 2025. [Paper](#)
24. "Exploring Emotional Nuances in Spoken Dialogue: Dataset Construction and Prediction of Emotional Dialogue Breakdown". *Hyuga Nakaguro et al.* IWSDS 2026. [Paper](#)
25. "DuplexChat: Constructing Speaker-Separated Full-Duplex Dialogue Speech at Scale for Spoken Dialogue Language Modeling". *Wataru Nakata et al.* arXiv 2026. [Paper](#)
26. "Dial HEALTHDIAL for Advice: A Multilingual and Multi-Parallel Spoken Dialogue Dataset for Knowledge-Grounded Information Seeking". *Songbo Hu et al.* ACL Findings 2026. [Paper](#)

## Evaluation & Benchmarks

### Evaluation Focus: Turn-taking & Interruption

1. "Investigating Speech Features for Continuous Turn-Taking Prediction Using LSTMs". *Matthew Roddy et al.* Interspeech 2018. [Paper](#)
2. "Turn-Taking Predictions Across Languages and Genres Using an LSTM Recurrent Neural Network". *Nigel G. Ward et al.* SLT 2018. [Paper](#)
3. "How Much Does Prosody Help Turn-taking? Investigations using Voice Activity Projection Models". *Erik Ekstedt et al.* SIGDIAL 2022. [Paper](#)
4. "Talking Turns: Benchmarking Audio Foundation Models on Turn-Taking Dynamics". *Siddhant Arora et al.* ICLR 2025. [Paper](#)
5. "Investigating the Impact of Incremental Processing and Voice Activity Projection on Spoken Dialogue Systems". *Yuya Chiba et al.* COLING 2025. [Paper](#)
6. "Synchronization and Turn-Taking in Full-Duplex Speech Dialogue Models". *Pablo Riera et al.* arXiv 2026. [Paper](#)
7. "Still Between Us? Evaluating and Improving Voice Assistant Robustness to Third-Party Interruptions". *Dongwook Lee et al.* ACL 2026. [Paper](#)
8. "Rethinking Binary Evaluation of Turn-Taking under Inherent Ambiguity". *Yunosuke Kubo et al.* SIGDIAL 2026. [Paper](#)

### Evaluation Focus: Full-duplex Interaction

1. "MMedFD: A Real-world Healthcare Benchmark for Multi-turn Full-Duplex Automatic Speech Recognition". *Hongzhao Chen et al.* arXiv 2025. [Paper](#)
2. "Full-Duplex-Bench: A Benchmark to Evaluate Full-duplex Spoken Dialogue Models on Turn-taking Capabilities". *Guan-Ting Lin et al.* arXiv 2025. [Paper](#)
3. "FD-Bench: A Full-Duplex Benchmarking Pipeline Designed for Full Duplex Spoken Dialogue Systems". *Yizhou Peng et al.* Interspeech 2025. [Paper](#)

4. "The ICASSP 2026 HumDial Challenge: Benchmarking Human-like Spoken Dialogue Systems in the LLM Era". *Zhixian Zhao et al.* ICASSP / arXiv 2026. [Paper](#)
5. "T-Voice: Benchmarking Full-Duplex Voice Agents on Real-World Domains". *Soham Ray et al.* ICML 2026. [Paper](#)
6. "Full-Duplex-Bench-v3: Benchmarking Tool Use for Full-Duplex Voice Agents Under Real-World Disfluency". *Guan-Ting Lin et al.* arXiv 2026. [Paper](#)
7. "Full-Duplex Interaction in Spoken Dialogue Systems: A Comprehensive Study from the ICASSP 2026 HumDial Challenge". *Chengyou Wang et al.* arXiv 2026. [Paper](#)
8. "Full-Duplex-Bench-v2: A Multi-Turn Evaluation Framework for Duplex Dialogue Systems with an Automated Examiner". *Guan-Ting Lin et al.* ACL 2026. [Paper](#)
9. "MTR-DuplexBench: Towards a Comprehensive Evaluation of Multi-Round Conversations for Full-Duplex Speech Language Models". *Zhang He et al.* ACL Findings 2026. [Paper](#)
10. "M3-DuplexBench: A Multi-Turn, Multilingual, Multidomain Benchmark for Full-Duplex Spoken Dialogue Models". *Ryo Fukuda et al.* arXiv 2026. [Paper](#)

### Evaluation Focus: Multi-turn Dialogue Capabilities

1. "SD-Eval: A Benchmark Dataset for Spoken Dialogue Understanding Beyond Words". *Junyi Ao et al.* NeurIPS 2024 (Datasets & Benchmarks). [Paper](#)
2. "Contextual Interactive Evaluation of TTS Models in Dialogue Systems". *Siyang Wang et al.* Interspeech 2024. [Paper](#)
3. "Benchmarking Open-ended Audio Dialogue Understanding for Large Audio-Language Models". *Kuofeng Gao et al.* arXiv 2024. [Paper](#)
4. "URO-Bench: Towards Comprehensive Evaluation for End-to-End Spoken Dialogue Models". *Ruiqi Yan et al.* EMNLP Findings 2025. [Paper](#)
5. "WavReward: Spoken Dialogue Models With Generalist Reward Evaluators". *Shengpeng Ji et al.* arXiv 2025. [Paper](#)
6. "MAD: A Benchmark for Multi-Turn Audio Dialogue Fact-Checking". *Chaewan Chun et al.* arXiv 2025. [Paper](#)
7. "VoxRole: A Comprehensive Benchmark for Evaluating Speech-Based Role-Playing Agents". *Weihao Wu et al.* arXiv 2025. [Paper](#)
8. "Evaluating Bias in Spoken Dialogue LLMs for Real-World Decisions and Recommendations (FairDialogue)". *Yihao Wu et al.* arXiv 2025. [Paper](#)
9. "MULTI-Bench: A Multi-turn Interactive Benchmark for Assessing Emotional Intelligence Ability of Spoken Dialogue Models". *Yayue Deng et al.* arXiv 2025. [Paper](#)
10. "Audio MultiChallenge: A Multi-Turn Evaluation of Spoken Dialogue Systems on Natural Human Interaction". *Advait Gosai et al.* arXiv 2025. [Paper](#)
11. "VoxDialogue: Can Spoken Dialogue Systems Understand Information Beyond Words?". *Xize Cheng et al.* ICLR 2025. [Paper](#)
12. "EchoMind: An Interrelated Multi-level Benchmark for Evaluating Empathetic Speech Language Models". *Li Zhou et al.* arXiv 2025. [Paper](#)
13. "SOVA-Bench: Benchmarking the Speech Conversation Ability for LLM-based Voice Assistant". *Yixuan Hou et al.* Interspeech 2025. [Paper](#)

14. "MTalk-Bench: Evaluating Speech-to-Speech Models in Multi-Turn Dialogues via Arena-style and Rubrics Protocols". *Yuhao Du et al.* arXiv 2025. [Paper](#)
15. "VoiceAgentBench: Are Voice Assistants Ready for Agentic Tasks?". *Dhruv Jain et al.* arXiv 2025. [Paper](#)
16. "C3: A Bilingual Benchmark for Spoken Dialogue Models Exploring Challenges in Complex Conversations". *Chengqian Ma et al.* EMNLP 2025. [Paper](#)
17. "HumDial-EIBench: A Human-Recorded Multi-Turn Emotional Intelligence Benchmark for Audio Language Models". *Shuiyuan Wang et al.* arXiv 2026. [Paper](#)
18. "VoiceBench: Benchmarking LLM-Based Voice Assistants". *Yiming Chen et al.* TACL 2026. [Paper](#)
19. "The Context Trap: Why End-to-End Audio Language Models Fail Multi-turn Dialogues". *Zhi Rui Tam et al.* IWSDS 2026. [Paper](#)
20. "SDiaReward: Modeling and Benchmarking Spoken Dialogue Rewards with Modality and Colloquialness". *Jingyu Lu et al.* ACL 2026. [Paper](#)
21. "SpeakerSleuth: Can Large Audio-Language Models Judge Speaker Consistency across Multi-turn Dialogues?". *Jonggeun Lee et al.* ACL 2026. [Paper](#)
22. "Style Amnesia: Investigating Speaking Style Degradation and Mitigation in Multi-Turn Spoken Language Models". *Yu-Xiang Lin et al.* ACL Findings 2026. [Paper](#)

## Models

### Interaction Mode: Turn-taking

1. "Improving End-of-Turn Detection in Spoken Dialogues by Detecting Speaker Intentions as a Secondary Task". *Zakaria Aldeneh et al.* ICASSP 2018. [Paper](#)
2. "Neural Dialogue Context Online End-of-Turn Detection". *Ryo Masumura et al.* SIGDIAL 2018. [Paper](#)
3. "Prediction of Turn-Taking Using Multitask Learning with Prediction of Backchannels and Fillers". *Kohei Hara et al.* Interspeech 2018. [Paper](#)
4. "Turn-Taking Prediction Based on Detection of Transition Relevance Place". *Kohei Hara et al.* Interspeech 2019. [Paper](#)
5. "TurnGPT: A Transformer-Based Language Model for Predicting Turn-Taking in Spoken Dialog". *Erik Ekstedt and Gabriel Skantze.* EMNLP Findings 2020. [Paper](#)
6. "Projection of Turn Completion in Incremental Spoken Dialogue Systems". *Erik Ekstedt and Gabriel Skantze.* SIGDIAL 2021. [Paper](#)
7. "Timing Generating Networks: Neural Network Based Precise Turn-Taking Timing Prediction in Multiparty Conversation". *Shinya Fujie et al.* Interspeech 2021. [Paper](#)
8. "Using Transition Duration to Improve Turn-taking in Conversational Agents". *Charles Threlkeld et al.* SIGDIAL 2022. [Paper](#)
9. "When can I Speak? Predicting Initiation Points for Spoken Dialogue Agents". *Siyan Li et al.* SIGDIAL 2022. [Paper](#)

10. "Voice Activity Projection: Self-supervised Learning of Turn-taking Events". *Erik Ekstedt and Gabriel Skantze* Interspeech 2022. [Paper](#)
11. "Gated Multimodal Fusion with Contrastive Learning for Turn-Taking Prediction in Human-Robot Dialogue". *Jiudong Yang et al.* ICASSP 2022. [Paper](#)
12. "Response Timing Estimation for Spoken Dialog System Using Dialog Act Estimation". *Jin Sakuma et al.* Interspeech 2022. [Paper](#)
13. "Turn-Taking Prediction for Natural Conversational Speech". *Shuo-Yiin Chang et al.* Interspeech 2022. [Paper](#)
14. "Generative Spoken Dialogue Language Modeling". *Tu Anh Nguyen et al.* arXiv 2022 / TACL 2023. [Paper](#)
15. "SpeechGPT: Empowering Large Language Models with Intrinsic Cross-Modal Conversational Abilities". *Dong Zhang et al.* EMNLP Findings 2023. [Paper](#)
16. "Let's Go Real Talk: Spoken Dialogue Model for Face-to-Face Conversation". *Se Park et al.* ACL 2024. [Paper](#)
17. "Qwen2-Audio Technical Report". *Yunfei Chu et al.* arXiv 2024. [Paper](#)
18. "Style-Talker: Finetuning Audio Language Model and Style-Based Text-to-Speech Model for Fast Spoken Dialogue Generation". *Yinghao Aaron Li et al.* arXiv 2024. [Paper](#)
19. "EMOVA: Empowering Language Models to See, Hear and Speak with Vivid Emotions". *Kai Chen et al.* arXiv 2024. [Paper](#)
20. "Internalizing ASR with Implicit Chain of Thought for Efficient Speech-to-Speech Conversational LLM". *Robin Shinghei Yuen et al.* NeurIPS 2024. [Paper](#)
21. "Building a Taiwanese Mandarin Spoken Language Model: A First Attempt". *Chih-Kai Yang et al.* arXiv 2024. [Paper](#)
22. "LLaMA-Omni: Seamless Speech Interaction with Large Language Models". *Qingkai Fang et al.* arXiv 2024. [Paper](#)
23. "Mini-Omni: Language Models Can Hear, Talk While Thinking in Streaming". *Zhifei Xie et al.* arXiv 2024. [Paper](#)
24. "GLM-4-Voice: Towards Intelligent and Human-Like End-to-End Spoken Chatbot". *Aohan Zeng et al.* arXiv 2024. [Paper](#)
25. "Triadic Multi-party Voice Activity Projection for Turn-taking in Spoken Dialogue Systems". *Mikey Elmers et al.* Interspeech 2025. [Paper](#)
26. "OpenOmni: Advancing Open-Source Omnimodal Large Language Models with Progressive Multimodal Alignment and Real-time Emotional Speech Synthesis". *Run Luo et al.* arXiv 2025. [Paper](#)
27. "GOAT-SLM: A Spoken Language Model with Paralinguistic and Speaker Characteristic Awareness". *Hongjie Chen et al.* arXiv 2025. [Paper](#)
28. "OpenS2S: Advancing Fully Open-Source End-to-End Empathetic Large Speech Language Model". *Chen Wang et al.* arXiv 2025. [Paper](#)
29. "Step-Audio 2 Technical Report". *StepFun Team et al.* arXiv 2025. [Paper](#)
30. "InteractiveOmni: A Unified Omni-modal Model for Audio-Visual Multi-turn Dialogue". *Wenwen Tong et al.* arXiv 2025. [Paper](#)



31. "Step-Audio: Unified Understanding and Generation in Intelligent Speech Interaction". *Ailin Huang et al.* arXiv 2025. [Paper](#)
32. "Baichuan-Audio: A Unified Framework for End-to-End Speech Interaction". *Tianpeng Li et al.* arXiv 2025. [Paper](#)
33. "Kimi-Audio Technical Report". *KimiTeam et al.* arXiv 2025. [Paper](#)
34. "Qwen3-Omni Technical Report". *Jin Xu et al.* arXiv 2025. [Paper](#)
35. "PACHAT: Persona-Aware Speech Assistant for Multi-party Dialogue". *Dongjie Fu et al.* EMNLP 2025. [Paper](#)
36. "VITA-Audio: Fast Interleaved Cross-Modal Token Generation for Efficient Large Speech-Language Model". *Zuwei Long et al.* arXiv 2025. [Paper](#)
37. "AV-Dialog: Spoken Dialogue Models with Audio-Visual Input". *Tuochao Chen et al.* arXiv 2025. [Paper](#)
38. "Qwen2.5-Omni Technical Report". *Qwen Team et al.* arXiv 2025. [Paper](#)
39. "LLaMA-Omni 2: LLM-based Real-time Spoken Chatbot with Autoregressive Streaming Speech Synthesis". *Qingkai Fang et al.* ACL 2025. [Paper](#)
40. "VoxMind: An End-to-End Agentic Spoken Dialogue System". *Tianle Liang et al.* ACL 2026. [Paper](#)
41. "ZipVoice-Dialog: Non-Autoregressive Spoken Dialogue Generation with Flow Matching". *Han Zhu et al.* ACL Findings 2026. [Paper](#)

### Interaction Mode: Half-duplex / Controlled Barge-in

1. "An Incremental Turn-Taking Model for Task-Oriented Dialog Systems". *Andrei C. Coman et al.* Interspeech 2019. [Paper](#)
2. "Oh, Jeez! or Uh-Huh? A Listener-Aware Backchannel Predictor on ASR Transcriptions". *Daniel Ortega et al.* ICASSP 2020. [Paper](#)
3. "BPM\_MT: Enhanced Backchannel Prediction Model Using Multi-Task Learning". *Jin Yea Jang et al.* EMNLP 2021. [Paper](#)
4. "Device Directedness with Contextual Cues for Spoken Dialog Systems". *Dhanush Bekal et al.* arXiv 2022. [Paper](#)
5. "Contextual Acoustic Barge-In Classification for Spoken Dialog Systems". *Dhanush Bekal et al.* Interspeech 2022. [Paper](#)
6. "Freeze-Omni: A Smart and Low Latency Speech-to-Speech Dialogue Model with Frozen LLM". *Xiong Wang et al.* arXiv 2024. [Paper](#)
7. "Mini-Omni2: Towards Open-source GPT-4o with Vision, Speech and Duplex Capabilities". *Zhifei Xie et al.* arXiv 2024. [Paper](#)
8. "VITA: Towards Open-Source Interactive Omni Multimodal LLM". *Chaoyou Fu et al.* arXiv 2024. [Paper](#)
9. "SHANKS: Simultaneous Hearing and Thinking for Spoken Language Models". *Cheng-Han Chiang et al.* arXiv 2025. [Paper](#)

### Interaction Mode: Full-duplex

1. "Duplex Conversation in Outbound Agent System". *Chunxiang Jin et al.* Interspeech 2021. [Paper](#)

2. "Duplex Conversation: Towards Human-like Interaction in Spoken Dialogue Systems". *Ting-En Lin et al.* arXiv 2022. [Paper](#)
3. "Beyond Turn-Based Interfaces: Synchronous LLMs as Full-Duplex Dialogue Agents". *Bandhav Veluri et al.* EMNLP 2024. [Paper](#)
4. "Moshi: A Speech-Text Foundation Model for Real-Time Dialogue". *Alexandre Defossez et al.* arXiv 2024. [Paper](#)
5. "SALMONN-omni: A Codec-free LLM for Full-duplex Speech Understanding and Generation". *Wenyi Yu et al.* arXiv 2024. [Paper](#)
6. "A Full-duplex Speech Dialogue Scheme Based On Large Language Models". *Peng Wang et al.* NeurIPS 2024. [Paper](#)
7. "MinMo: A Multimodal Large Language Model for Seamless Voice Interaction". *FunAudioLLM Team et al.* arXiv 2025. [Paper](#)
8. "SALM-Duplex: Efficient and Direct Duplex Modeling for Speech-to-Speech Language Model". *Ke Hu et al.* arXiv 2025. [Paper](#)
9. "OmniFlatten: An End-to-end GPT Model for Seamless Voice Conversation". *Qinglin Zhang et al.* ACL 2025. [Paper](#)
10. "Towards a Japanese Full-duplex Spoken Dialogue System". *Atsumoto Ohashi et al.* Interspeech 2025. [Paper](#)
11. "FireRedChat: A Pluggable, Full-Duplex Voice Interaction System with Cascaded and Semi-Cascaded Implementations". *Junjie Chen et al.* arXiv 2025. [Paper](#)
12. "Incorporating Dialogue State Tracking into Japanese Full-duplex Task-oriented Spoken Dialogue Model". *Yuya Chiba et al.* IJCNLP-AACL 2025. [Paper](#)
13. "Chronological Thinking in Full-Duplex Spoken Dialogue Language Models". *Donghang Wu et al.* arXiv 2025. [Paper](#)
14. "LLM-Enhanced Dialogue Management for Full-Duplex Spoken Dialogue Systems". *Hao Zhang et al.* arXiv 2025. [Paper](#)
15. "Language Model Can Listen While Speaking". *Ziyang Ma et al.* AACL 2025. [Paper](#)
16. "FlashLabs Chroma 1.0: A Real-Time End-to-End Spoken Dialogue Model with Personalized Voice Cloning". *Tanyu Chen et al.* arXiv 2026. [Paper](#)
17. "PersonaPlex: Voice and Role Control for Full Duplex Conversational Speech Models". *Rajarshi Roy et al.* arXiv 2026. [Paper](#)
18. "MiniCPM-o 4.5: Towards Real-Time Full-Duplex Omni-Modal Interaction". *Junbo Cui et al.* arXiv 2026. [Paper](#)
19. "DuplexSLA: A Full-Duplex Spoken Language Model with Synchronized Speech, Language, and Action". *Haoyang Zhang et al.* arXiv 2026. [Paper](#)
20. "BayLing-Duplex: Native Full-Duplex Speech Dialogue with a Single Autoregressive LLM". *Qingkai Fang et al.* arXiv 2026. [Paper](#)
21. "F-Actor: Controllable Conversational Behavior in Full-Duplex Models". *Maike Züfle et al.* ACL Findings 2026. [Paper](#)
22. "Hierarchical Acoustic-Semantic Modeling: Modality Separation and Semantic Coherence for Full-Duplex SLMs". *Zhenyu Liu et al.* ACL 2026. [Paper](#)



23. "JoyAI-Talker: Full-Duplex Speech Interactive Large Model Built for Empathetic Voice Agents". *Yinhao Bai et al.* arXiv 2026. [Paper](#)
24. "SoulX-Duplug: Plug-and-Play Streaming State Prediction Module for Realtime Full-Duplex Speech Conversation". *Ruiqi Yan et al.* arXiv 2026. [Paper](#)

## Training Methods

### Interaction Mode: Turn-taking

1. "Towards Generalized Models for Task-Oriented Dialogue Modeling on Spoken Conversations". *Ruijie Yan et al.* arXiv 2022. [Paper](#)
2. "Adapting Text-based Dialogue State Tracker for Spoken Dialogues". *Jaeseok Yoon et al.* DSTC 2023. [Paper](#)
3. "SpeechAlign: Aligning Speech Generation to Human Preferences". *Dong Zhang et al.* NeurIPS 2024. [Paper](#)
4. "Data-Centric Improvements for Enhancing Multi-Modal Understanding in Spoken Conversation Modeling (ASK-QA)". *Maximillian Chen et al.* ACL Findings 2025. [Paper](#)
5. "WavRAG: Audio-Integrated Retrieval Augmented Generation for Spoken Dialogue Models". *Yifu Chen et al.* ACL 2025. [Paper](#)
6. "Enhancing Speech-to-Speech Dialogue Modeling with End-to-End Retrieval-Augmented Generation". *Pengchao Feng et al.* EMNLP Findings 2025. [Paper](#)
7. "Aligning Spoken Dialogue Models from User Interactions". *Anne Wu et al.* arXiv 2025. [Paper](#)
8. "Chain-of-Thought Training for Open E2E Spoken Dialogue Systems". *Siddhant Arora et al.* arXiv 2025. [Paper](#)
9. "VoiceTextBlender: Augmenting Large Language Models with Speech Capabilities via Single-Stage Joint Speech-Text Supervised Fine-Tuning". *Yifan Peng et al.* NAACL 2025. [Paper](#)
10. "Approaching Dialogue State Tracking via Aligning Speech Encoders and LLMs". *Šimon Sedláček et al.* arXiv 2025. [Paper](#)
11. "The Speech-LLM Takes It All: A Truly Fully End-to-End Spoken Dialogue State Tracking Approach". *Nizar El Ghazal et al.* arXiv 2025. [Paper](#)
12. "Optimizing Conversational Quality in Spoken Dialogue Systems with Reinforcement Learning from AI Feedback". *Siddhant Arora et al.* arXiv 2026. [Paper](#)
13. "WavAlign: Enhancing Intelligence and Expressiveness in Spoken Dialogue Models via Adaptive Hybrid Post-Training". *Yifu Chen et al.* ACL Findings 2026. [Paper](#)

### Interaction Mode: Half-duplex / Streaming

1. "A Methodology for Turn-Taking Capabilities Enhancement in Spoken Dialogue Systems Using Reinforcement Learning". *Hatim Khouzaimi et al.* Computer Speech & Language 2018. [Paper](#)
2. "Stream RAG: Instant and Accurate Spoken Dialogue Systems with Streaming Tool Usage". *Siddhant Arora et al.* arXiv 2025. [Paper](#)
3. "Dual-Reasoner: Bridging Interleaved Atomicity and Streaming Latency via Thinking-while-Talking". *Yangzhuo Li et al.* ACL Findings 2026. [Paper](#)

## Interaction Mode: Full-duplex

1. "FLM-Audio: Natural Monologues Improves Native Full-Duplex Chatbots via Dual Training". *Yiqun Yao et al.* arXiv 2025. [Paper](#)
2. "Reinforcement Learning Enhanced Full-Duplex Spoken Dialogue Language Models for Conversational Interactions". *Chen Chen et al.* COLM 2025. [Paper](#)
3. "Multi-Faceted Interactivity Alignment in Full-Duplex Speech Models". *Atsumoto Ohashi et al.* arXiv 2026. [Paper](#)
4. "Reproducing Proficiency-Conditioned Dialogue Features with Full-duplex Spoken Dialogue Models". *Takao Obi et al.* IWSDS 2026. [Paper](#)
5. "Effects of Dialogue Corpora Properties on Fine-Tuning a Moshi-Based Spoken Dialogue Model". *Yuto Abe et al.* IWSDS 2026. [Paper](#)
6. "Dual-Axis Generative Reward Model Toward Semantic and Turn-taking Robustness in Interactive Spoken Dialogue Models". *Yifu Chen et al.* ACL 2026. [Paper](#)
7. "Sommelier: Scalable Open Multi-turn Audio Pre-processing for Full-duplex Speech Language Models". *Kyudan Jung et al.* ACL Industry Track 2026. [Paper](#)

## Survey Coverage Comparison

**Table 1. Coverage comparison of surveys related to Multi-Turn Speech Dialogue, including fine-grained organization aligned with this collection.**

General Surveys are listed first, followed by core surveys. ✓ = systematic coverage; △ = partial or supporting coverage; - = not a substantive focus. Interaction Taxonomy denotes an explicit taxonomy of conversational interaction. Fine-grained Organization assesses whether a survey structures its analysis using the evaluation focuses, model interaction modes, and training interaction modes adopted in this collection; relevant discussion without a matching subdivision is marked △.

[illegible]

# System and Model Comparison

Table 2. Comparison of all papers listed under Models for Multi-Turn Speech Dialogue.

✓ = explicitly reported support; △ = partial, asymmetric, or controller-mediated support; - = not reported, not applicable, or not a primary capability. Method names link to the papers.

| Method                                              | Scope     | Interaction Capabilities |          |            |                       | Modalities    |        | System Architecture |            |            | Speech Interface      | Acoustic / Speech Model                                       | Release |
|-----------------------------------------------------|-----------|--------------------------|----------|------------|-----------------------|---------------|--------|---------------------|------------|------------|-----------------------|---------------------------------------------------------------|---------|
|                                                     |           | Streaming                | Barge-in | Concurrent | Backchannel / Overlap | Speech Output | Vision | Cascaded            | Semi-casc. | End-to-end |                       |                                                               |         |
| Interaction Mode: Turn-taking                       |           |                          |          |            |                       |               |        |                     |            |            |                       |                                                               |         |
| EOT + Intent                                        | Component | △                        | -        | -          | -                     | -             | -      | -                   | -          | -          | Acoustic features     | BiLSTM acoustic EOT/intent classifier                         | 2018.04 |
| Neural Online EOT                                   | Component | ✓                        | -        | -          | -                     | -             | -      | -                   | -          | -          | Acoustic + context    | LSTM online EOT predictor (no generator)                      | 2018.07 |
| MTL Turn-taking                                     | Component | △                        | -        | -          | ✓                     | -             | -      | -                   | -          | -          | Acoustic features     | Multitask LSTM turn/BC/filler predictor                       | 2018.09 |
| TRP Turn-taking                                     | Component | △                        | -        | -          | -                     | -             | -      | -                   | -          | -          | Speech features       | LSTM TRP predictor (no generator)                             | 2019.09 |
| TurnGPT                                             | Component | △                        | -        | -          | -                     | -             | -      | -                   | -          | -          | Text tokens           | GPT-2 turn-shift LM (text-only)                               | 2020.11 |
| Turn Completion Projection                          | Component | ✓                        | -        | -          | -                     | -             | -      | -                   | -          | -          | Incremental text      | Incremental completion predictor (no generator)               | 2021.07 |
| Timing Generating Networks                          | Component | △                        | -        | -          | -                     | -             | -      | -                   | -          | -          | Timing features       | LSTM timing-generating network                                | 2021.08 |
| Transition Duration                                 | Component | △                        | -        | -          | -                     | -             | -      | -                   | -          | -          | Timing features       | Transition-duration predictor (no generator)                  | 2022.09 |
| Initiation Point Predictor                          | Component | △                        | -        | -          | -                     | -             | -      | -                   | -          | -          | Speech + context      | Neural initiation-point predictor (no generator)              | 2022.09 |
| VAP                                                 | Component | ✓                        | -        | -          | ✓                     | -             | -      | -                   | -          | -          | Waveform              | Causal Transformer voice-activity predictor                   | 2022.09 |
| Gated Multimodal Fusion                             | Component | △                        | -        | -          | -                     | -             | ✓      | -                   | -          | -          | Audio-visual features | Gated audio-visual turn predictor                             | 2022.05 |
| Response Timing Estimator                           | Component | △                        | -        | -          | -                     | -             | -      | -                   | -          | -          | Dialogue acts         | DNN response-timing estimator                                 | 2022.09 |
| Natural TT Predictor                                | Component | △                        | -        | -          | -                     | -             | -      | -                   | -          | -          | Conversational speech | Neural turn-taking predictor (no generator)                   | 2022.09 |
| dGSLM                                               | Model     | △                        | -        | △          | ✓                     | ✓             | -      | -                   | -          | ✓          | HuBERT units          | HuBERT units + dual-stream Transformer + HiFi-GAN             | 2022.03 |
| SpeechGPT                                           | Model     | -                        | -        | -          | -                     | ✓             | -      | -                   | -          | ✓          | HuBERT units          | HuBERT units + LLaMA + Unit HiFi-GAN                          | 2023.05 |
| RealTalk                                            | System    | △                        | -        | -          | -                     | ✓             | -      | -                   | -          | ✓          | Speech features       | SSL speech encoder + dialogue Transformer + unit decoder      | 2024.08 |
| Qwen2-Audio                                         | Model     | -                        | -        | -          | -                     | -             | -      | ✓                   | -          | -          | Whisper features      | Whisper-large-v3 encoder + Qwen-7B                            | 2024.07 |
| Style-Talker                                        | System    | △                        | -        | -          | -                     | ✓             | -      | -                   | ✓          | -          | Audio LM + TTS        | Speech encoder + audio LM + style-conditioned TTS             | 2024.08 |
| EMOVA                                               | Model     | -                        | -        | -          | -                     | ✓             | ✓      | -                   | -          | ✓          | Semantic + acoustic   | Semantic-acoustic tokenizer + LLM + style decoder             | 2024.09 |
| IntrinsicVoice                                      | Model     | △                        | -        | -          | -                     | ✓             | -      | -                   | -          | ✓          | HuBERT units          | HuBERT units + LLaMA + GroupFormer decoder                    | 2024.09 |
| Taiwanese Mandarin SLM                              | Model     | -                        | -        | -          | -                     | ✓             | -      | -                   | -          | ✓          | Discrete units        | Discrete-unit Transformer LM + unit vocoder                   | 2024.11 |
| LLaMA-Omni                                          | System    | △                        | -        | -          | -                     | ✓             | -      | -                   | -          | ✓          | Whisper + units       | Whisper encoder + LLaMA + streaming unit decoder              | 2024.09 |
| Mini-Omni                                           | System    | △                        | -        | -          | -                     | ✓             | -      | -                   | -          | ✓          | SNAC codec            | Whisper encoder + Qwen + SNAC token decoder                   | 2024.08 |
| GLM-4-Voice                                         | System    | ✓                        | -        | -          | -                     | ✓             | -      | -                   | -          | ✓          | Speech tokenizer      | Speech tokenizer + GLM-4 + flow-matching decoder              | 2024.12 |
| Triadic VAP                                         | Component | ✓                        | -        | -          | ✓                     | -             | -      | -                   | -          | -          | Multi-party waveform  | Causal Transformer triadic VAP predictor                      | 2025.08 |
| OpenOmni                                            | Model     | △                        | -        | -          | -                     | ✓             | ✓      | -                   | -          | ✓          | Speech tokens         | Pretrained speech encoder + LLM + lightweight speech decoder  | 2025.01 |
| GOAT-SLM                                            | Model     | -                        | -        | -          | -                     | ✓             | -      | -                   | -          | ✓          | Speech tokens         | Dual semantic/acoustic heads + speech decoder                 | 2025.07 |
| OpenS2S                                             | Model     | -                        | -        | -          | -                     | ✓             | -      | -                   | -          | ✓          | Speech tokens         | BLSP-Emo + streaming interleaved speech decoder               | 2025.07 |
| Step-Audio 2                                        | System    | ✓                        | -        | -          | -                     | ✓             | -      | -                   | -          | ✓          | Speech-text tokens    | Step-Audio tokenizer + LLM + speech-token decoder             | 2025.07 |
| InteractiveOmni                                     | Model     | △                        | -        | -          | -                     | ✓             | ✓      | -                   | -          | ✓          | Audio-visual tokens   | Audio-visual encoders + unified Transformer decoder           | 2025.10 |
| Step-Audio                                          | System    | ✓                        | -        | -          | -                     | ✓             | -      | -                   | -          | ✓          | Speech-text tokens    | Step-Audio tokenizer + 130B LLM + speech decoder              | 2025.02 |
| Baichuan-Audio                                      | System    | ✓                        | -        | -          | -                     | ✓             | -      | -                   | -          | ✓          | Multi-codebook        | Speech encoder + Baichuan LLM + multi-codebook decoder        | 2025.02 |
| Kimi-Audio                                          | Model     | ✓                        | -        | -          | -                     | ✓             | -      | -                   | -          | ✓          | Continuous + tokens   | Audio encoder/tokenizer + Kimi LLM + speech decoder           | 2025.04 |
| Qwen3-Omni                                          | Model     | ✓                        | -        | -          | -                     | ✓             | ✓      | -                   | -          | ✓          | Continuous + codec    | Qwen Thinker + autoregressive Talker decoder                  | 2025.09 |
| PACHAT                                              | System    | △                        | -        | -          | ✓                     | ✓             | -      | -                   | ✓          | -          | Speech frontend + TTS | Speech frontend + persona-aware LLM + expressive TTS          | 2025.11 |
| VITA-Audio                                          | Model     | ✓                        | -        | -          | -                     | ✓             | -      | -                   | -          | ✓          | Interleaved tokens    | Audio encoder + LLM + interleaved speech-token decoder        | 2025.05 |
| AV-Dialog                                           | Model     | -                        | -        | -          | -                     | ✓             | ✓      | -                   | -          | ✓          | Audio-visual features | Audio-visual encoders + dialogue Transformer + speech decoder | 2025.11 |
| Qwen2.5-Omni                                        | Model     | ✓                        | -        | -          | -                     | ✓             | ✓      | -                   | -          | ✓          | Continuous + codec    | Qwen Thinker + speech-token Talker + vocoder                  | 2025.03 |
| LLaMA-Omni 2                                        | System    | ✓                        | -        | -          | -                     | ✓             | -      | -                   | -          | ✓          | Discrete units        | Audio encoder + LLaMA + streaming unit decoder                | 2025.05 |
| VoxMind                                             | System    | △                        | -        | -          | -                     | ✓             | -      | -                   | -          | ✓          | Speech tokens         | Speech tokenizer + agentic dialogue LM + speech decoder       | 2026.07 |
| ZipVoice-Dialog                                     | Model     | △                        | -        | -          | -                     | ✓             | -      | -                   | -          | ✓          | Continuous latent     | Flow-matching DiT speech generator                            | 2026.07 |
| Interaction Mode: Half-duplex / Controlled Barge-in |           |                          |          |            |                       |               |        |                     |            |            |                       |                                                               |         |
| Incremental TT Model                                | Component | ✓                        | -        | -          | -                     | -             | -      | -                   | -          | -          | Incremental features  | Incremental neural TT predictor (no generator)                | 2019.09 |
| Listener-aware BC                                   | Component | △                        | -        | -          | ✓                     | -             | -      | -                   | -          | -          | ASR text              | Listener-aware neural BC predictor (no generator)             | 2020.05 |
| BPM_MIT                                             | Component | △                        | -        | -          | ✓                     | -             | -      | -                   | -          | -          | Speech + text         | Multitask neural BC predictor (no generator)                  | 2021.11 |
| Device Directedness                                 | Component | △                        | -        | -          | -                     | -             | -      | -                   | -          | -          | Acoustic + context    | Acoustic-context directedness classifier                      | 2022.11 |
| Acoustic Barge-in                                   | Component | △                        | ✓        | -          | -                     | -             | -      | -                   | -          | -          | Acoustic + context    | Contextual acoustic barge-in classifier                       | 2022.09 |
| Freeze-Omni                                         | System    | ✓                        | ✓        | △          | △                     | ✓             | -      | -                   | -          | ✓          | Continuous + units    | Speech encoder + frozen LLM + chunkwise unit decoder          | 2024.11 |
| Mini-Omni2                                          | System    | ✓                        | ✓        | △          | △                     | ✓             | ✓      | -                   | -          | ✓          | SNAC codec            | Whisper encoder + Qwen + SNAC token decoder                   | 2024.10 |
| VITA                                                | System    | ✓                        | ✓        | △          | △                     | ✓             | ✓      | -                   | ✓          | -          | Encoder + TTS         | Audio encoder + LLM + streaming TTS                           | 2024.08 |
| SHANKS                                              | Add-on    | ✓                        | △        | △          | -                     | ✓             | -      | -                   | ✓          | -          | Streaming features    | Streaming speech encoder + side-path state predictor          | 2025.10 |
| Interaction Mode: Full-duplex                       |           |                          |          |            |                       |               |        |                     |            |            |                       |                                                               |         |
| Duplex Conversation (Outbound)                      | System    | ✓                        | ✓        | ✓          | △                     | ✓             | -      | ✓                   | -          | -          | ASR / TTS             | Streaming ASR + dialogue controller + TTS                     | 2021.08 |
| Duplex Conversation                                 | System    | ✓                        | ✓        | ✓          | ✓                     | ✓             | -      | -                   | ✓          | -          | ASR / TTS             | Two-channel interaction model + ASR/TTS                       | 2022.05 |
| Synchronous LLM                                     | Model     | ✓                        | ✓        | ✓          | ✓                     | ✓             | -      | -                   | -          | ✓          | Speech units          | Dual-channel speech units + synchronous LLaMA                 | 2024.09 |
| Moshi                                               | Model     | ✓                        | ✓        | ✓          | ✓                     | ✓             | -      | -                   | -          | ✓          | Mimi codec            | Mimi codec + Helium temporal/depth Transformers               | 2024.10 |
| SALMONN-omni                                        | Model     | ✓                        | ✓        | ✓          | ✓                     | ✓             | -      | -                   | -          | ✓          | Continuous embeddings | Speech encoder + codec-free asynchronous speech decoder       | 2024.11 |
| Full-duplex LLM Scheme                              | System    | ✓                        | ✓        | ✓          | △                     | ✓             | -      | ✓                   | -          | -          | ASR / LLM / TTS       | Streaming ASR + LLM dialogue manager + TTS                    | 2024.05 |
| MinMo                                               | Model     | ✓                        | ✓        | ✓          | ✓                     | ✓             | ✓      | -                   | -          | ✓          | Speech tokens         | Speech encoder/tokenizer + 8B LLM + speech decoder            | 2025.01 |
| SALM-Duplex                                         | Model     | ✓                        | ✓        | ✓          | △                     | ✓             | -      | -                   | -          | ✓          | Encoder + codec       | Streaming speech encoder + LLM + codec decoder                | 2025.05 |
| OmniFlatten                                         | Model     | ✓                        | ✓        | ✓          | ✓                     | ✓             | -      | -                   | -          | ✓          | Speech-text tokens    | Flattened speech-text streams + decoder-only Transformer      | 2025.07 |
| Japanese FD System                                  | System    | ✓                        | ✓        | ✓          | △                     | ✓             | -      | ✓                   | -          | -          | ASR / LLM / TTS       | Streaming ASR + LLM dialogue manager + TTS                    | 2025.08 |
| FireRedChat                                         | System    | ✓                        | ✓        | ✓          | △                     | ✓             | -      | ✓                   | ✓          | -          | ASR / TTS or codec    | ASR/TTS cascade or codec-based semi-cascade                   | 2025.09 |
| FD Dialogue + DST                                   | System    | ✓                        | ✓        | ✓          | △                     | ✓             | -      | ✓                   | ✓          | -          | ASR / DST / TTS       | Streaming ASR + DST/dialogue manager + TTS                    | 2025.12 |
| Chronological Thinking                              | Model     | ✓                        | ✓        | ✓          | ✓                     | ✓             | -      | -                   | -          | ✓          | Synchronized tokens   | Synchronized speech tokens + chronological Transformer        | 2025.10 |
| LLM-Enhanced FD-DM                                  | System    | ✓                        | ✓        | ✓          | △                     | ✓             | -      | ✓                   | -          | -          | ASR / LLM-DM / TTS    | Streaming ASR + LLM-enhanced dialogue manager + TTS           | 2025.02 |
| LSLM                                                | Model     | ✓                        | ✓        | ✓          | △                     | ✓             | -      | -                   | -          | ✓          | SSL + speech tokens   | SSL speech encoder + channel-fusion LLM + token decoder       | 2025.02 |
| Chroma 1.0                                          | System    | ✓                        | ✓        | ✓          | △                     | ✓             | -      | -                   | -          | ✓          | Speech tokens         | Neural codec + real-time dialogue Transformer                 | 2026.01 |
| PersonaPlex                                         | Model     | ✓                        | ✓        | ✓          | △                     | ✓             | -      | -                   | -          | ✓          | Neural codec          | Mimi codec + duplex Transformer + voice conditioning          | 2026.02 |
| MiniCPM-o 4.5                                       | Model     | ✓                        | ✓        | ✓          | △                     | ✓             | ✓      | -                   | -          | ✓          | Omni-modal stream     | Omni encoder + MiniCPM LLM + streaming speech decoder         | 2026.04 |
| DuplexSLA                                           | Model     | ✓                        | ✓        | ✓          | ✓                     | ✓             | -      | -                   | -          | ✓          | Continuous + discrete | Speech encoder + joint language/action LM + codec decoder     | 2026.05 |
| Bayling-Duplex                                      | Model     | ✓                        | ✓        | ✓          | ✓                     | ✓             | -      | -                   | -          | ✓          | Audio tokens          | Audio tokenizer + autoregressive duplex LLM + decoder         | 2026.06 |
| F-Actor                                             | Add-on    | ✓                        | ✓        | ✓          | ✓                     | ✓             | -      | -                   | ✓          | -          | Control states        | Behavior-state controller (uses host speech model)            | 2026.07 |
| Hierarchical A-S Modeling                           | Model     | ✓                        | ✓        | ✓          | ✓                     | ✓             | -      | -                   | -          | ✓          | Semantic + acoustic   | Hierarchical semantic/acoustic duplex Transformer             | 2026.07 |
| JoyAI-Talker                                        | Model     | ✓                        | ✓        | ✓          | △                     | ✓             | -      | -                   | -          | ✓          | Semantic + acoustic   | Semantic/acoustic tokenizers + dual-level Transformer         | 2026.08 |
| SoulX-Duplug                                        | Add-on    | ✓                        | ✓        | ✓          | △                     | ✓             | -      | -                   | ✓          | -          | Streaming ASR         | Streaming ASR + semantic-state predictor (host TTS)           | 2026.03 |

Abbreviations: ASR, automatic speech recognition; TTS, text-to-speech; SSL, self-supervised learning; LLM, large language model; VAD, voice activity detection. Acoustic / Speech Model reports the principal speech encoder, tokenizer, dialogue backbone, decoder, or controller evaluated by each paper; Release uses YYYY.MM and follows the cited archival publication month, or the first arXiv month when the cited work is a preprint.

# Development Trajectory: From Spoken Dialogue Models to Voice Agents

Positioning the starting point. The engineering precursor appeared in duplex task systems, but this collection treats dGSLM (March 2022) as the opening of the generative spoken-dialogue-model track because it directly learns textless, two-channel dialogue generation. The subsequent mainline is a progressive coupling of speech representation, interaction timing, low-latency duplex generation, external tools, and explicit reasoning/action.

| Phase                           | Period     | Representative milestones                                                                                  | Technical shift                                                                                                                             | Capability boundary                                                                                                                     |
|---------------------------------|------------|------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Engineering precursor           | 2021       | <a href="#">Duplex Conversation in Outbound Agent System</a>                                               | Modular duplex control demonstrates interruption-aware task delivery before a unified generative dialogue model exists.                     | Controller-centered task completion; speech understanding, dialogue policy, and synthesis remain separately orchestrated.               |
| Model-track opening             | 2022.03-05 | <a href="#">Generative Spoken Dialogue Language Modeling (dGSLM)</a> ; <a href="#">Duplex Conversation</a> | Textless dual-channel generation and learned conversational timing move the field from pipeline behavior toward dialogue modeling.          | Naturalistic speech and turn-taking emerge, but instruction following, external knowledge, and tool use remain limited.                 |
| Speech-native foundation models | 2023.05    | <a href="#">SpeechGPT</a>                                                                                  | Discrete speech units, an LLM backbone, and cross-modal instruction tuning unify speech perception and generation.                          | Instruction-following speech interaction becomes feasible, while most systems still depend on turn-based exchanges.                     |
| Real-time full-duplex models    | 2024.09-10 | <a href="#">Synchronous LLMs</a> ; <a href="#">Moshi</a> ; <a href="#">OmniFlatten</a>                     | Streaming token generation, parallel user/system streams, and staged half-to-full-duplex training remove explicit turn boundaries.          | Low-latency overlap, interruption, and backchannels improve naturalness; external action and long-horizon planning are not yet central. |
| Tool-augmented voice assistants | 2025.10    | <a href="#">Stream RAG</a> ; <a href="#">VoiceAgentBench</a>                                               | Speech-in/speech-out systems issue tool calls during incoming speech, while benchmarks introduce multi-tool and multi-turn agentic tasks.   | Grounded task execution emerges, but robust orchestration and long-horizon planning remain open problems.                               |
| Agentic spoken dialogue         | 2026.04-07 | <a href="#">Full-Duplex-Bench-v3</a> ; <a href="#">DuplexSLA</a> ; <a href="#">VoxMind</a>                 | Think-before-speak reasoning, dynamic tool management, synchronized language/action generation, and realistic tool-use evaluation converge. | The reason-plan-act loop is integrated with speech; persistent memory, safety, and reliable long-horizon autonomy remain unresolved.    |

## The Mainline

- Representation: ASR/text/TTS pipelines -> discrete speech units -> parallel semantic and acoustic streams.
- Interaction: explicit turn boundaries -> streaming and controlled barge-in -> full-duplex overlap and backchannels.
- Capability: dialogue generation -> cross-modal instruction following -> retrieval and tool use -> reason-plan-act behavior.
- Evaluation: speech quality and turn timing -> latency/interruption -> multi-tool correctness and end-to-end task completion.

Scope note. All representative milestones above are already listed in the earlier Datasets, Evaluation & Benchmarks, Models, or Training Methods sections; no additional bibliography entry was introduced solely for this trajectory.

# Development Roadmap and Capability Curve

The main curve summarizes a qualitative progression from generative spoken-dialogue modeling to integrated spoken agents. Year branches expand representative works from this collection: color denotes the interaction or agency path, while dashed tags identify benchmark or training evidence. The vertical progression is not a benchmark score.

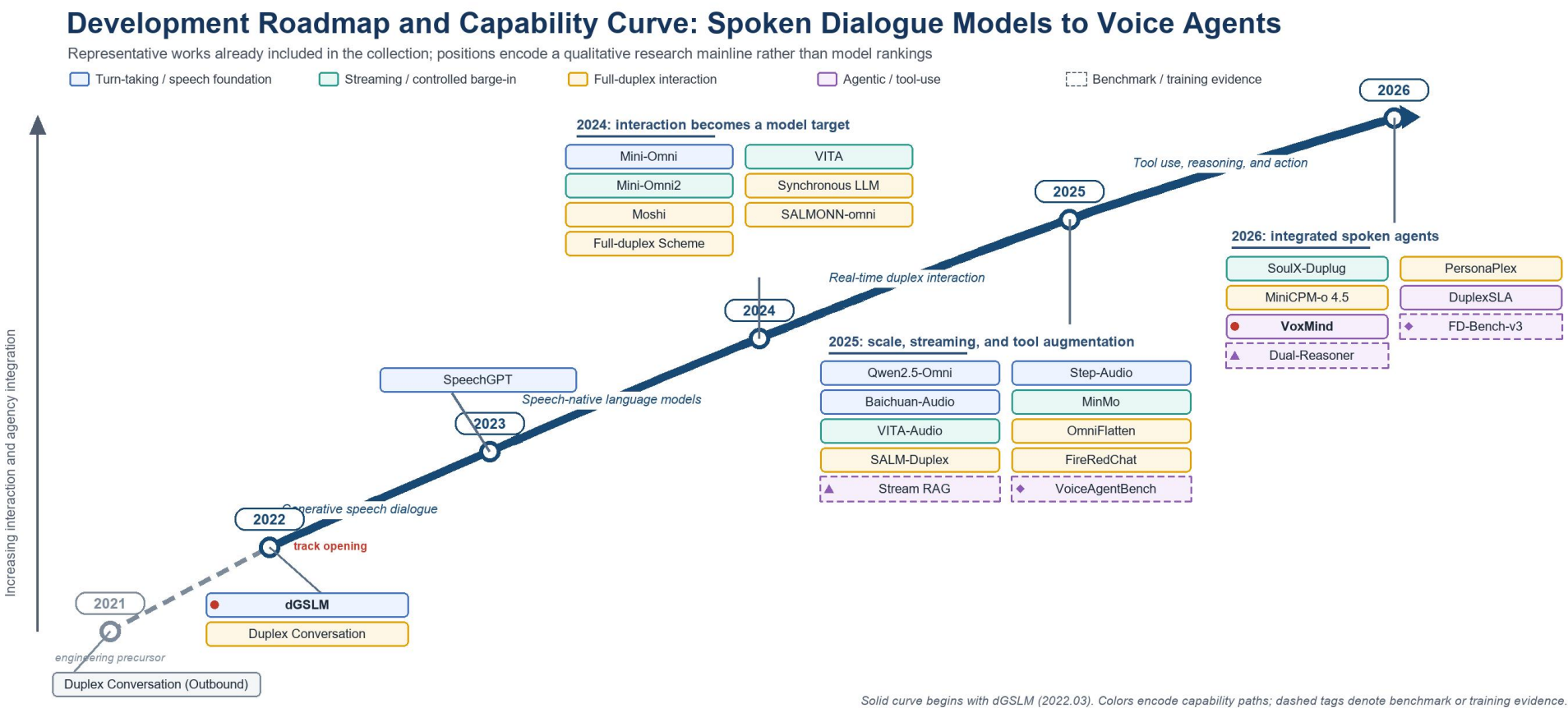


Figure 1. Academic roadmap and conceptual capability curve for Multi-Turn Speech Dialogue. The dotted 2021 segment provides engineering context; the solid model-development mainline begins with dGSLM in March 2022. Branches show representative works already included in this collection.

Interpretation. The field evolves along three coupled axes: speech-native representation, increasingly synchronous interaction, and growing task agency. The dense expansion after 2024 reflects diversification rather than a single winning architecture; by 2025-2026, retrieval, tool use, reasoning, action generation, and agent-oriented evaluation begin to close the loop from model-centric conversation to task-capable voice agents.